Encoding Arithmetic In First Order Logic (DRAFT)

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1 Introduction

The purpose of this document is to provide an intermediate representation (IR) for encoding arithmetic (i.e. $+, \times, =, <, >$) using first order logic (FOL) without the equality symbol $=$ or free variables. Furthermore, a first order logic statement of the form $\exists a \forall x \exists y \forall z \forall z'\ P(a, x, y, z, z')$ where P is quantifier free and only contains binary primitive predicates.

2 Overview

We construct a first order sentence (yet to be constructed) in the language of First Order Logic without equality and without bounded variables and without an arbitrary domain where the satisfying model is an IR of addition and multiplication (to do). We start out with a sentence of the form $\forall x \forall z \forall z' \in \mathbb{N}\ P(x, z, z', z'', z''')$ where P is quantifier free, equality is allowed, and we can reference arbitrary successors (i.e. $x+1$, $z+1$, $z'+1$,...) and prove the correctness for the IR of addition. Afterwards, we prove the correctness of the IR for multiplication, before eliminating certain quantifiers, replacing equality symbol with auxillary predicates, and replacing successors with predicates and/or treating them as Skolemized existential variables. The final result is a first order sentence of the form $\exists a \forall x \exists y \forall z \forall z'\ P(a, x, y, z, z')$.

2.1 Construction Idea

The basic idea is to construct a model using only binary primitive predicates, and of the form $\forall \exists \forall *$ (signature of the first order sentence). Natively, addition and multiplication equations lend themselves to ternary predicate representations:

$$\text{Add}(a, b, c) \iff a + b = c$$

$$\text{Mult}(a, b, d) \iff a \times b = d$$

However, we cannot treat these ternary predicates as primitives per our problem statement. The idea instead is to look at the arguments of a binary predicates not as operands, but as ways to store data.
In particular, we treat the columns of a binary predicate (second argument) as a data index to retrieve information about addition and multiplication operations.
Every addition rule $a + b = c$ should be stored in some column n. Similarly, every multiplication rule $a \times b = d$ should also be stored in some column m. For convenience and efficiency, we can use the same column n to store the operands of both addition and multiplication. We introduce binary predicates A, B, C, D for this.

$$\mathrm{Add}(a, b, c) \iff (\exists n : A(a, n) \wedge B(b, n) \wedge C(c, n))$$

$$\mathrm{Mult}(a, b, d) \iff (\exists n : A(a, n) \wedge B(b, n) \wedge D(d, n))$$

So far, this guarantees every addition and multiplication operation can be "stored" in some respective column. Ideally, we would simply want to "compute" addition and multiplication by addressing the operands (a, b) and getting the result (i.e. $c = a+b$, $d = a \times b$). The simplest idea to enforce this is to ensure that every column has at least some stored value of a, b, c, and d. Furthermore, to prevent conflict and simplify the proof of correctness, we will want to assert that such a, b, c, d are unqiue. We can write this as :

$$\forall n, \exists_1 a, b, c \ A(a, n) \wedge B(b, n) \wedge C(c, n) \wedge D(d, n)$$

($\exists_1$ is the uniqueness quantifier)

This way, every $n \in \mathbb{N}$ corresponds to exactly one addition and multiplication rule with the same operands a and b.

3 Arithmetic in FOL without equality

The idea of the previous section was to store an infinite "additive" and "multiplicative" "lookup table" for addition and multiplication. However, we will need to construct an explicit mapping that can map operands (a, b) to a column index n. Furthermore, first order statement of the type AEA* naturally only allow one successor reference (i.e. $x \to x + 1$) by performing Skolemization on the existential quantifier. From a geometric standpoint, we can interpret the model structure of AEA* as behaving like a 2-dimensional grid in the first quadrant of the coordinate plane. The columns represent horizontal movements, and the rows represent vertical movements.

With this in mind, we want to construct a function, or more specifically, a bijection (since $\mathbb{N}^{\nvDash}$ and $\mathbb{N}$ are both countable) where shifting (± 1) the operands moves to the next column (to the right) as much as possible. Let f define the mapping of a, b to its index column n.

3.1 Bijections

Let $f : \mathbb{N}^2 \to \mathbb{N}$ be a function from the set $\mathbb{N}^2$ to $\mathbb{N}$ defined by:

$$f(a,b) = \binom{a+b+1}{2} = \frac{(a+b)^2 + (a+b)}{2} + a = \frac{a^2 + 2ab + b^2 + a + b}{2} + a$$

The first few mappings defined by f are:

$$\begin{aligned}
f(0,0) &: 0 \\
f(0,1) &: 1 \\
f(1,0) &: 2 \\
f(0,2) &: 3 \\
f(1,1) &: 4 \\
f(2,0) &: 5 \\
f(0,3) &: 6 \\
f(1,2) &: 7 \\
f(2,1) &: 8 \\
f(3,0) &: 9 \\
f(0,4) &: 10
\end{aligned}$$

etc...

Intuitively, this appears to be a bijection from $\mathbb{N}^{\nvDash} \to \mathbb{N}$. In fact we will prove this because this is a crucial step in proving the correctness of an IR model of addition and multiplication that is constructed later.

Theorem 3.1. *f is a bijective function from $\mathbb{N}^2$ to $\mathbb{N}$.*

Proof. To show f is a bijection from $\mathbb{N}^2$ to $\mathbb{N}$. We must show that:

- f is onto (all natural numbers have the form $(a^2 + 2ab + b^2 + a + b)/2 + a$)
- f is one-to-one (all distinct pairs (a, b) map to different values)

We will show that f is onto.

By the well ordering principle, for every natural number n, there exists a smallest k such that

$$n < \binom{k+1}{2}$$

. Because k is the smallest integer such that $n < \binom{k+1}{2}$, it must be the case that $\binom{k}{2} \leq n$.

Because $\binom{k}{2} \leq n$, there is a non-negative integer r such that $\binom{k}{2} + r = n$.

Since $0 \leq n < \binom{k+1}{2}$, we have:

$$0 < \binom{k+1}{2}$$

$$0 < \frac{k^2+k}{2}$$

$$0 < k^2 + k$$

$$-k < k^2$$

If k is a natural number, then the equality holds for all $k > 0$. Hence, $k \geq 1$.
We have from earlier:
$\binom{k}{2} + r = n$
So

$$\binom{k}{2} + r < \binom{k+1}{2}$$

$$r < \binom{k+1}{2} - \binom{k}{2}$$

$$r < k$$

(which follows Pascal's Identity)

Because a and b were defined to be arbitrary natural numbers, and $k \geq 1$, we can assert the existence of some a and b such that $a+b+1 = k$. Because r was an integer less than k (or $\leq k-1$), we have $r < a+b+1$, or $r \leq a+b$. Since a and b are arbitrary numbers that sum to $k-1$, and $r \leq k-1$, we can, without loss of generality, let $r = a$.
Hence, we have shown the existence of integers a and b such that for any n, $n = \binom{a+b+1}{2} + a$.

We now need to show that for any pairs of integers (a, b) and (c, d), $(f(a, b) = f(c, d)) \implies (a = c) \wedge (b = d)$ (i.e. injectivity).
Suppose that $f(a, b) = f(c, d) = n \in \mathbb{N}$.

We have $\binom{a+b+1}{2} + a = \binom{c+d+1}{2} + c$.

Let $u = a + b + 1$ and $v = c + d + 1$. We have:

$$\binom{u}{2} + a = \binom{v}{2} + c$$

Without loss of generality (by the well ordering principle), suppose that $u \leq v$. Let $u + w = v$ for some nonnegative integer w.

We have $a < u$ and $u \geq 1$. This, along with Pascal's Triangle Identity (which implies $\binom{u+1}{2} > \binom{u}{2}$ since $u > 0$), yields:

$$\binom{u}{2} + a \leq \binom{u}{2} + a + b \; (= \binom{u}{2} + u - 1) < \binom{u+1}{2} \leq \binom{u+i}{2}$$

$$\text{(for any } i > 0)$$

Since $u + w = v$ and $\binom{u}{2} + a = \binom{u+w}{2} + a$, but $\binom{u}{2} + a < \binom{u+i}{2} + a$ for $i > 0$, w must be 0, so $u = v$, and $a + b = c + d$.
Hence,

$$\binom{u}{2} + a = \binom{v}{2} + c = n \implies a = c$$

and

$$a + b = c + d \implies b = d$$

Since f is both onto and one-to-one or injective, f must be a bijection from $\mathbb{N}^2$ to $\mathbb{N}$. $\square$

4 IR of Addition in First Order Logic

To encode addition rules using binary predicates, we rely on the construction methods described in Section 2, in conjunction with the bijection f we outlined in Section 3. The idea is to use diagonal top-down (for the first operand a) and bottom-up (for the second operand b) encodings, to translate between different addition rules between successive columns (i.e. n to $n+1$). When the bottom up rule (i.e. b) approaches zero (the first row), we then "slide" the sum c to the next value $c + 1$ and reset the top down counter a to zero, while setting b to $c + 1$, the the top-down and bottom-up loops repeat. Because a starts at 0 and traverses to c, and b starts at c and traverses to zero, and c consistently always shifts by one, every summation rule $a + b = c$ is uniquely addressable in a certain column. For instance, Column 0 might address the rule $0 + 0 = 0$ while column 4 might address the rule $1 + 1 = 2$.

4.1 The First Order Statement

Statement 4.1.

$$\forall x \forall z \forall z' \in \mathbb{N}:$$

$$\begin{aligned}
A(0,0) &\wedge \\
B(0,0) &\wedge \\
C(0,0) &\wedge \\
((A(z,x) \wedge A(w,x)) \implies z = z') &\wedge \\
((B(z,x) \wedge B(w,x)) \implies z = z') &\wedge \\
((C(z,x) \wedge C(w,x)) \implies z = z') &\wedge \\
(C(x,z) \implies (C(x,z+1) \vee C(x+1,z+1))) &\wedge \\
((C(x,z) \wedge A(x,z)) \implies \neg A(x+1,z+1)) &\wedge \\
((\neg C(x,z) \wedge A(x,z)) \implies A(x+1,z+1)) &\wedge \\
(B(x+1,z) \implies B(x,z+1)) &\wedge \\
(B(0,x) \implies A(0,x+1)) &\wedge \\
((B(0,x) \wedge C(z,x)) \implies \neg C(z,x+1)) &\wedge \\
((\neg B(0,x) \wedge C(z,x)) \implies C(z,x+1)) &\wedge \\
((A(0,x) \wedge C(z,x)) \implies B(z,x)) &\wedge
\end{aligned}$$

Theorem 4.1. *For all natural numbers n, there exists natural numbers a, b, c such that $a+b=c$ and $A(a,n) \wedge B(b,n) \wedge C(c,n)$. Furthermore, such a, b, c are unique.*

Proof. The theorem must be proven for all eligible pairs (n, a, b, c). To make the proof more efficient, we invoke the at-most-one rules in Statement 4.1, effectively showing the uniqueness of a, b, and c such that $A(a,n) \wedge B(b,n) \wedge C(c,n)$:

$$\begin{aligned}
((A(z,n) \wedge A(z',n)) \implies z = z') &\wedge \\
((B(z,n) \wedge B(z',n)) \implies z = z') &\wedge \\
((C(z,n) \wedge C(z',n)) \implies z = z') &
\end{aligned}$$

We can prove the existence criteria (that a, b, c actually exist), and the sum relationship ($a+b=c$) by induction. The remainder of the proof proceeds by induction on n. Suppose $S(n)$ is the statement we want to prove for all $n \in \mathbb{N}$:

$$S(n) =$$

$$\exists a, b, c : A(a,n) \wedge B(b,n) \wedge C(c,n)$$

and

$$(A(a,n) \wedge B(b,n) \wedge C(c,n)) \implies (a+b=c)$$

The initial case $S(0)$ is trivially met by the axioms of Statement 4.1 (where $n, a, b, c = 0$):

$$A(0,0) \wedge B(0,0) \wedge C(0,0)$$

and

$$a+b=c \text{ is true : } 0+0=0$$

For the induction on n, Suppose that $S(n)$ is true. Let (a, b, c) be the unique pair satisfying $S(n)$ [inductive hypothesis]. Let (a', b', c') be the unique pair satisfying $S(n+1)$, if it exists. We divide the proof into two cases based on whether or not $B(0,n)$ is true or false.

First assume $B(0,n)$ is false. Because we have the existence of a b such that $B(b,n)$ it must be the case that $b > 0$, and so the rule (letting $x+1=b$):

$$B(b,z) \implies B(b-1, z+1)$$

applies (as $b-1$ is nonnegative). So $B(b-1, n+1)$ is true.
Additionally (from the falsity of $B(0,x)$), the rule

$$((\neg B(0,n) \wedge C(c,n)) \implies C(c, n+1))$$

also applies (letting $x=n$ and $z=c$). Hence, $C(c, n+1)$ is true.

Now letting $x=a$ and $z=n$, we obtain the following two implications:

$$((C(a,n) \wedge A(a,n)) \implies \neg A(a+1, n+1))$$

$$((\neg C(a,n) \wedge A(a,n)) \implies A(a+1, n+1))$$

If $C(a,n)$ were true, then $c=a$ by uniqueness constraints, and $a+b=c=a$, which requires $b=0$, which contradicts that $b>0$. Therefore, $C(a,n)$ must be false (and $a \neq c$), in which case, the second implication rule applies, while the first is vacuously true. By the second rule, we have $A(a+1, n+1)$ to be true. By letting $a' = a+1$, $b' = b-1$, $c' = c$ have found a pair (a', b', c') such that:

$$A(a', n+1) \wedge B(b', n+1) \wedge C(c', n+1)$$

and

$$a' + b' = c' \text{ is true : } a' + b' = a + 1 + b - 1 = a + b = c = c'$$

From the uniqueness of a', b', c', no such other pairs exist and the statement S holds for $n+1$ when $B(0, n)$ is false.

Now assume, for the second case, that $B(0, n)$ is true. In that case, it follows directly from the rule

$$B(0, n) \implies A(0, n+1)$$

in Statement 4.1 that $A(0, n+1)$ is also true. By the rule (substituting $x = n$ and $z = c$ in the original statement):

$$(B(0, n) \land C(c, n)) \implies \neg C(c, n+1)$$

and by the rule

$$C(c, n) \implies (C(c, n+1) \lor C(c+1, n+1))$$

we can conclude that since $C(c, n+1)$ is false (first rule listed above), $C(c+1, n+1)$ must be true (second rule listed above). Using the rule ($z = c+1$, $x = n+1$):

$$(A(0, n+1) \land C(c+1, n+1)) \implies B(c+1, n+1)$$

in conjunction with the previously stated two, we can conclude that $B(c+1, n+1)$ is true. Hence we have the existence of elements $a' = 0$, $b' = c+1$, and $c' = c+1$ such that:

$$A(a', n+1) \land B(b', n+1) \land C(c', n+1)$$

and

$$a' + b' = c' \text{ is true : } 0 + (c+1) = c+1$$

By the at most one constraint of a', b', c', $S(n+1)$ must therefore be true (since there are no other (a', b', c') pairs to prove).
Therefore, since the statement $S(n+1)$ is true in both cases (depending on whether $B(0, n)$ is true or false), it holds. $\square$

Theorem 4.2. *For all natural numbers n, a, b, $A(a, n)$ and $B(b, n)$ are true if and only if $n = f(a, b)$.*

Proof. In the previous theorem, we established that there exists a unique a and b such $A(a, n)$ and $B(b, n)$ are true, for every n. By Theorem 3.1, because f is a bijection, there must exist unique u and v such $n = f(u, v)$. The goal is to show that $u = a$ and $v = b$.
We can prove this by induction:

$$S(n) = \{\forall a, b, u, v \in \mathbb{N}\, ((f(u, v) = n) \land A(a, n) \land B(b, n)) \implies ((u = a) \land (v = b))\}$$

For $S(0)$, we previously established that $(a,b,c) = (0,0,0)$ is the unique pair (a,b,c).
On the other hand if $f(u,v) = \binom{u+v+1}{2} + u = \frac{u^2+2uv+v^2+u+v}{2} + u$ is zero, then $u = 0$ and $v^2 + v = 0 \implies v = 0$. Hence $u = a = 0$ and $v = b = 0$.
For the inductive step, we can, similar to the previous theorem, split the step into two cases.
The inductive hypothesis assumes that $u = b$ and $v = a$.
As with the previous proof let (a,b,u,v) be the pair that satisfies $S(n)$. Let $c = a+b$. Let (a',b') be the pair satisfying $A(a',n+1) \wedge B(b',n+1)$, $c' = a'+b'$, and (u',v') be the pair satisfying $f(u',v') = n+1$. Both (a',b') and (u',v') must exist by Theorems 4.1 and 3.1, respectively. The objective is to show they are equal.

Case I: $b \neq 0$.
In this case, since b is a natural number, $b > 0$. It was shown in Theorem 4.1 that for $n+1$, both $A(a+1,n+1)$ and $B(b-1,n+1)$ are both true. Hence $a' = a+1$ and $b' = b+1$. Because of the uniqueness of solutions to $f(u',v') = n+1$, it is sufficient to show that $f(a+1,b-1) = n$. (which would imply that $u = a' = a+1$ and $v = b' = b-1$.
Using the binomial form:

$$f(a+1,b-1) = \binom{a+1+b-1+1}{2} + a + 1 = \binom{a+b+1}{2} + a + 1 =$$

$$(\binom{a+b+1}{2} + a) + 1 =$$

$$n+1$$

Hence, since $f(a+1,b-1) = n+1$, $u = a' = a+1$ and $v = b' = b-1$ and the induction step is proven.

Case II : $b = 0$.
In this case, we have $f(a,0) = \binom{a+1}{2} + a = \frac{a^2+a}{2} + a = \frac{a^2+3a}{2} = n$. From the inductive reasoning of Theorem 4.1, it was shown that for $n+1$, $A(0,n+1)$ and $B(c+1,n+1)$, and consequently $C(c+1,n+1)$ is true $(a' = 0, b' = c+1, c' = c+1)$. As $f(u',v') = n+1$, it is sufficient to show that $f(0,c+1) = f(0,a+b+1)$ evaluates to $n+1$:

$$f(0,c+1) = \binom{0+(c+1)+1}{2} + 0 = \binom{c+2}{2}$$

Yet by the induction Hypothesis, $f(a,b) = \binom{a+b+1}{2} + a = \binom{c+1}{2} + a = n$.

Using Pascal's Identity, we have:

$$\binom{c+2}{2} - \binom{c+1}{2} = c + 1 = a + b + 1$$

So we can express

$$\binom{c+2}{2} =$$

$$\binom{c+1}{2} + c + 1 =$$

$$\binom{a+b+1}{2} + a + b + 1 =$$

$$\binom{a+1}{2} + a + 1 =$$

$$\text{(because } b = 0\text{)}$$

$$(\binom{a+1}{2} + a) + 1 =$$

$$f(a,b) + 1 =$$

$$n + 1$$

$$\text{(applying the inductive hypothesis)}$$

Hence, $f(0, c+1) = \binom{c+2}{2} = n + 1$, so $u' = a' = 0$ and $v' = b' = c + 1$.
In either case $(b \neq 0)$ or $b = 0$, the inductive step holds and $S(n+1)$ is true.

$\square$

Corollary 4.1. *For all natural numbers a, b, there exist exactly one natural number n such that $A(a, n)$, $B(b, n)$, and consequently, $C(a + b, n)$ are all true.*

This directly follows from the reasoning Theorem 4.1 to show they exist, the reasoning of Theorem 4.2 to establish the relationship of a, b to f, and Theorems 3.1 and 4.1 to show that such a and b are unique, and again, Theorem 4.1 to show that if $C(c, n)$ is true, $c = a + b$.

5 IR of Multiplication in First Order Logic

In the addition representation, we were able to diagonally "shift" the operands a and b in opposite directions to enumerate all integers a, b that sum to any given integer c, and when the enumeration was done (i.e. $b = 0$), we increment c and repeat the process forever.
Unfortunately, multiplicative properties are difficult to enumerate in a sequential column fashion. Instead, now that we have additive properties established for A, B, and C, we can use those to encode multiplication only using recursive

addition rules. The idea is to use the fact that $a \times b$ is equivalent to adding a, b number of times. The recursive solution is as follows:

$$\begin{cases} a \times b = 0, & \text{if } b = 0 \\ a \times b = a \times (b-1) + a, & \text{if } b > 0 \end{cases}$$

To implement this behavior in first order logic, we objectively want to encode the property

$$(A(a,n) \wedge B(b,n) \wedge D(d,n)) \implies a \times b = d$$

By Theorem 4.2, we already have the operands a and b encoded into a unique column n (and their sum c), so the only remaining objective is to encode their product d, into the same column. As with A, B, C, we will want to establish uniqueness for D. We will also need an intermediate variable P to represent the previous product (i.e. $p = a \times (b-1)$) and then address the column which computes the sum of a and p, and then assert that the product of a and b is equal to the sum of a and p. We describe the first order statement below. We use z_2, z_3, z_4 as extra quantifiers instead of the extra z' quantifier (variable naming convention change). The conjunctions contained in the first set of square brackets are the addition rules we proved hold in Section 4. The conjunctions contained in the second set of square brackets are the multiplication rules we intend to encode and prove hold as intended.

5.1 The First Order Statement

Statement 5.1.

$$\forall x \forall z \forall z_2 \forall z_3 \forall z_4 \in \mathbb{N}:$$

$$
\begin{aligned}
&[A(0,0) \wedge \\
&B(0,0) \wedge \\
&C(0,0) \wedge \\
&((A(z,x) \wedge A(z_2,x)) \implies z = z_2) \wedge \\
&((B(z,x) \wedge B(z_2,x)) \implies z = z_2) \wedge \\
&((C(z,x) \wedge C(z_2,x)) \implies z = z_2) \wedge \\
&(C(x,z) \implies (C(x,z+1) \vee C(x+1,z+1))) \wedge \\
&((C(x,z) \wedge A(x,z)) \implies \neg A(x+1,z+1)) \wedge \\
&((\neg C(x,z) \wedge A(x,z)) \implies A(x+1,z+1)) \wedge \\
&(B(x+1,z) \implies B(x,z+1)) \wedge \\
&(B(0,x) \implies A(0,x+1)) \wedge \\
&((B(0,x) \wedge C(z,x)) \implies \neg C(z,x+1)) \wedge \\
&((\neg B(0,x) \wedge C(z,x)) \implies C(z,x+1)) \wedge \\
&((A(0,x) \wedge C(z,x)) \implies B(z,x))] \\
&\wedge \\
&[((D(z,x) \wedge D(z_2,x)) \implies z = z_2) \wedge \\
&((P(z,x) \wedge P(z_2,x)) \implies z = z_2) \wedge \\
&(B(0,x) \implies D(0,x)) \wedge \\
&(B(0,x) \implies \neg P(z,x)) \wedge \\
&((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4)) \wedge \\
&((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))]
\end{aligned}
$$

Theorem 5.1. *For all natural numbers n, there exists natural numbers a, b, d such that $a \times b = d$ and $A(a,n) \wedge B(b,n) \wedge D(d,n)$. Furthermore, such a, b, d are unique.*

Proof. From Theorem 4.1, we already know there exists a and b such that $A(a,n) \wedge B(b,n)$ are both true. Therefore, the only remaining objective is to show that $D(d,n)$ is true iff $a \times b = d$.
It helps to establish the uniqueness rules for D and P ahead of time:

$$((D(z,x) \wedge D(z_2,x)) \implies z = z_2)$$

$$((P(z,x) \wedge P(z_2,x)) \implies z = z_2)$$

Hence, we now only have to show that for this specific d, it exists, and is the product of a and b.
Rather than inducting in the columns as we did for the proof of theorem 4.1, we will instead induct on the value of b, treating a as an arbitrary fixed integer. Let S be the following statement:

$$S(b) : (\forall a \forall d\ (A(a,n) \wedge B(b,n) \wedge D(d,n)) \implies (a \times b = d))$$

S(0) reduces down to $(\forall a \forall d\ (A(a,n) \wedge B(0,n) \wedge D(d,n)) \implies (a \times 0 = d))$ so for the statement to hold, $d = 0$. We have the implication rule

$$(B(0,n) \implies D(0,n))$$

(substituting $x = n$) so for the implication to follow, $d = 0$ (which is true if $b = 0$) and the consequent $a \times 0 = 0$ is true, so $S(0)$ is true. If $d \neq 0$, then $D(d,n)$ is false and $S(0)$ vacuously true. In either case, $S(0)$ is true.
Now assume $S(b)$ is true. For a fixed $b > 0$ (we just showed $b = 0$ yields a true statement), we have the rule

$$((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4))$$

Making the substitutions $z = a$, $x = b$, $z_2 = d$, $z_3 = n$, $z_4 = m$, this becomes:

$$((A(a,n) \wedge B(b,n) \wedge D(d,n) \wedge A(a,m) \wedge B(b+1,m)) \implies P(d,m))$$

By the induction hypothesis, $a \times b = d$. The unique column m exists operands a and $b+1$, and we directly assert $P(a \times b, m)$ to be true. Since true values of P (for each column) are unique, we know that P uniquely encodes the previous product into the row addressing the successor of the second operand $b+1$. By the rule

$$((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))$$

Making the substitutions $x = a$, $z = d$, $z_2 = m$, $z_3 = m'$, $z_4 = a + d$, this becomes:

$$((A(a,m) \wedge P(d,m) \wedge A(a,m') \wedge B(d,m') \wedge C(a+d,m')) \implies D(a+d,m))$$

From the previous rule and deduction steps, $P(d,n)$ was implied to be true, $A(a,m)$ was true by assumption, there is a unique column m' containing operands a and d, respectively, and their sum $a+d$ (predicate C). Hence $D(a+d,m)$ is true.
Because m was assumed to be the unique column variable holding a and $b+1$ as operands (i.e. $A(a,m)$ and $B(b+1,m)$), and $a + d = a + a \times b = a \times (b+1)$, it follows that the statement $S(b+1)$ is true (since the value of d' such that $D(d',m)$ is true, is the product of a and $b+1$).

$\square$

In later sections [TODO], we will describe how to rewrite statement 5.1 using less quantifiers (quantifier elimination), at the cost of only adding a few more intermediate predicates, eliminating equality comparisons (uniqueness constraints), eliminating successor references to $z+1$, and eliminating the use of constant 0 (other than $(0,0)$).

6 Equality, Successor, and Constant Eliminations

In this section, we turn our attention to the statements that make use of equality, successor references (other than x), and constant zero (other than $(0,0)$). When we mention our First-Order statement of interest, we will be referring to Statement 5.1.

The uniqueness constraints we imposed are:

$$\begin{aligned}
(A(z,x) \wedge A(z_2,x)) &\implies z = z_2\\
(B(z,x) \wedge B(z_2,x)) &\implies z = z_2\\
(C(z,x) \wedge C(z_2,x)) &\implies z = z_2\\
(D(z,x) \wedge D(z_2,x)) &\implies z = z_2\\
(P(z,x) \wedge P(z_2,x)) &\implies z = z_2
\end{aligned}$$

The statements making references to successor $z+1$ are:

$$\begin{aligned}
C(x,z) &\implies (C(x,z+1) \vee C(x+1,z+1))\\
(C(x,z) \wedge A(x,z)) &\implies \neg A(x+1,z+1)\\
(\neg C(x,z) \wedge A(x,z)) &\implies A(x+1,z+1)\\
B(x+1,z) &\implies B(x,z+1)
\end{aligned}$$

The statements that make use of the constant 0 other than the point of origin are:

$$\begin{aligned}
B(0,x) &\implies A(0,x+1)\\
(B(0,x) \wedge C(z,x)) &\implies \neg C(z,x+1)\\
(\neg B(0,x) \wedge C(z,x)) &\implies C(z,x+1)\\
(A(0,x) \wedge C(z,x)) &\implies B(z,x)\\
B(0,x) &\implies D(0,x)\\
B(0,x) &\implies \neg P(z,x)
\end{aligned}$$

There are a few reasons why we want to re-write such statements that avoid using certain constants or symbols.
From an objective standpoint, the quantifier prefix form AEA* ($\forall\exists\forall*$) only allows for at most one successor reference, that is, Skolemnizing the existential quantifier and binding it as a function of the first universal statement. The structure does not permit that for all universal variables. Hence, we must get rid of terms like $z+1$ or find another way to preserve the structure.
Furthermore, equality is strictly forbidden in the classical version Entscheidungsproblem [REF NEEDED!!!]. The only reason equality is needed is to enforce uniqueness of values of A, B, C, D and P for each column, so enforcing the uniqueness constraint of say, row values of a such that $A(a, n)$ is true, but $a \neq a' \implies \neg A(a', n)$, is sufficient enough.
The reason we want to eliminate the constant zero (other than $(0, 0)$) is that the AEA* prefix does not allow for constant references. While in theory, we may assert that we have a non-empty domain via EAEA*, the first existential only serves that purpose, and is not intended to be used elsewhere other than fixing points at the origin.

6.1 Equality elimination

In this subsection, we will turn our attention to the elimination of the equality variable $=$ in statements like

$$(A(z, x) \wedge A(z_2, x)) \implies z = z_2$$

Before explaining the solution, it makes sense to first explain the general idea. Suppose we have a unary predicate $\mathbb{P}$ (not to confuse with P). The standard way to express that any true value for $\mathbb{P}$ is:

$$\forall x, z : (\mathbb{P}(x) \wedge \mathbb{P}(z)) \implies (x = z)$$

This is also what is called as an "at most one" constraint, because the statement is still true if no value of x makes $\mathbb{P}$ true, and true if there is exactly one value of x that makes $\mathbb{P}$ true.
To bypass using the equality sign, we introduce an auxiliary predicate $\mathbb{P}'$, which acts as a "switch" that toggles when the first value of x such that $\mathbb{P}(x)$ is true occurs.

$$\forall x, z : \neg(\mathbb{P}(x) \wedge \mathbb{P}'(x)) \wedge (\mathbb{P}(x) \implies \mathbb{P}'(x+1)) \wedge (\mathbb{P}'(x) \implies \mathbb{P}'(x+1))$$

By forcing $\mathbb{P}$ and $\mathbb{P}'$ to be mutually exclusive, and because every $n > x$ has $\mathbb{P}'(n)$ to be true, it forces $\mathbb{P}(n)$ to be false, meaning once $\mathbb{P}(x)$ occurs as a true value for some x, it can never occur again (at most one constraint). Alternatively, one can count "downwards" (towards zero) and achieve the same result (the same reasoning works, except $n < x$):

$$\forall x, z : \neg(\mathbb{P}(x) \wedge \mathbb{P}'(x)) \wedge (\mathbb{P}(x+1) \implies \mathbb{P}'(x)) \wedge (\mathbb{P}'(x+1) \implies \mathbb{P}'(x))$$

We may wish to use the latter set of implications for most cases, because it will make things more efficient and compact later on.

6.2 Successor elimination

In the previous section, we showed how we can eliminate equality symbols by enforcing uniqueness through auxiliary "switch" predicates.
Enforcing the successor behavior (specifically targeting $z+1$) without directly referencing it requires more auxiliary predicates to complete a two-step process. Initially, we have statements such as

$$C(x, z) \implies (C(x, z+1) \vee C(x+1, z+1))$$

that we want to re-write without the $z+1$ reference. We explain below how to eliminate that reference. Suppose we have a coordinate (x, z) and we want to "travel" to (or more technically, reference), $(x+1, z+1)$. We cannot reference that coordinate in one shot while not referencing $z+1$. Instead, we replace the reference mapping:

$$(x, z) \to (x+1, z+1)$$

with:

$$(x, z) \to (x+1, z)$$
$$(x+1, z) \to (x+1, z+1)$$

However, rather than writing the second leap as

$$(x+1, z) \to (x+1, z+1)$$

, we can simply write it as

$$(z, x) \to (z, x+1)$$

because we are universally quantifying over x and z.
Like the previous section, we will use $\mathbb{P}$ to explain the process, though we will treat $\mathbb{P}(x)$ as a binary predicate instead. Suppose we have:

$$\forall x, z : \mathbb{P}(x, z) \implies \mathbb{P}(x+1, z+1)$$

We will introduce the auxiliary predicate $\mathbb{P}^*$ to refer to the connecting step (in the two step process) to shift the second argument by one, and then finally, the first argument by one.

$$\forall x, z : (\mathbb{P}(z, x) \implies \mathbb{P}^*(z, x+1)) \ \wedge \ (\mathbb{P}^*(x, z) \implies \mathbb{P}(x+1, z))$$

Because x, z are universal, we can plug in, say, if $\mathbb{P}(u, v)$ is true, $x = v$ and $z = u$ which leads to $\mathbb{P}^*(u, v+1)$ from the first implication, and then plug in $x = u$ and $z = v+1$, to get $\mathbb{P}(u+1, v+1)$ from the second implication. It is, of course, worth noting that the same approach works in the opposite direction. For example, if we had instead

$$\forall x, z : \mathbb{P}(x+1, z+1) \implies \mathbb{P}(x, z)$$

we would use

$$\forall x, z : (\mathbb{P}(z, x+1) \implies \mathbb{P}^*(z, x)) \ \wedge \ (\mathbb{P}^*(x+1, z) \implies \mathbb{P}(x, z))$$

Here, we shifted to the immediate predecessor column in the first implication, and then with the second implication, apply that new truth to shift to the immediate predecessor row.

6.3 Non Origin Constant elimination

In this section, we intend to remove or rewrite references to zero such as

$$B(0, x) \implies A(0, x+1)$$

that are not the point of origin (i.e. like $A(0, 0)$). The solution to rewriting something like $B(0, x) \implies A(0, x+1)$ without 0 as the first argument is to create a binary predicate Z which is true iff the first argument is zero.

For example, if we had something like :

$$\forall x : \mathbb{P}(0, x)$$

we introduce a predicate Z such that:

$$\forall x, z : Z(0, 0) \wedge \neg Z(x+1, z) \wedge (Z(z, x) \implies Z(z, x+1))$$

This way, Z is true only along the first row (because it is false all successor rows, and true values propagate along rows where it is true once, i.e. at the origin, it propagates horizontally). We can directly replace the constant zero (with predicate $\mathbb{P}$ as):

$$\forall x, z : Z(z, x) \implies \mathbb{P}(z, x)$$

The implication only yields a consequence if the first argument (i.e. z) is zero, preserving logical equivalence of $\mathbb{P}$.

6.4 Rewriting the Arithmetic Sentence

In this second, we apply the structural equivalents discussed in Sections 6.1,6.2, and 6.3 to the sentence described in 5.1.

7 References

[TODO] @bookRogers1987, author = Rogers, Hartley, title = Theory of Recursive Functions and Effective Computability, publisher = MIT Press, year = 1987, address = Cambridge, MA